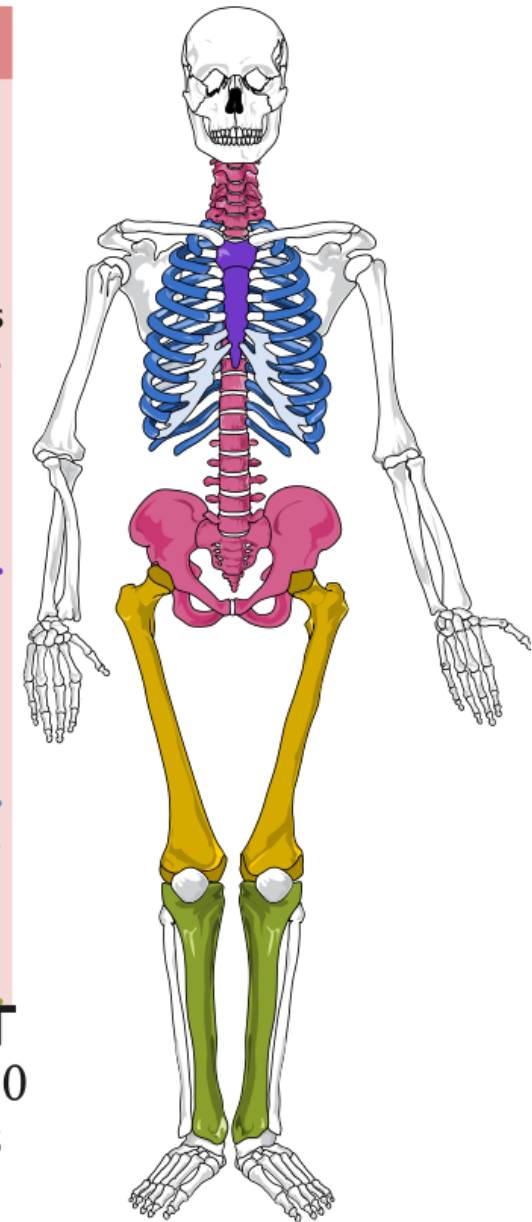
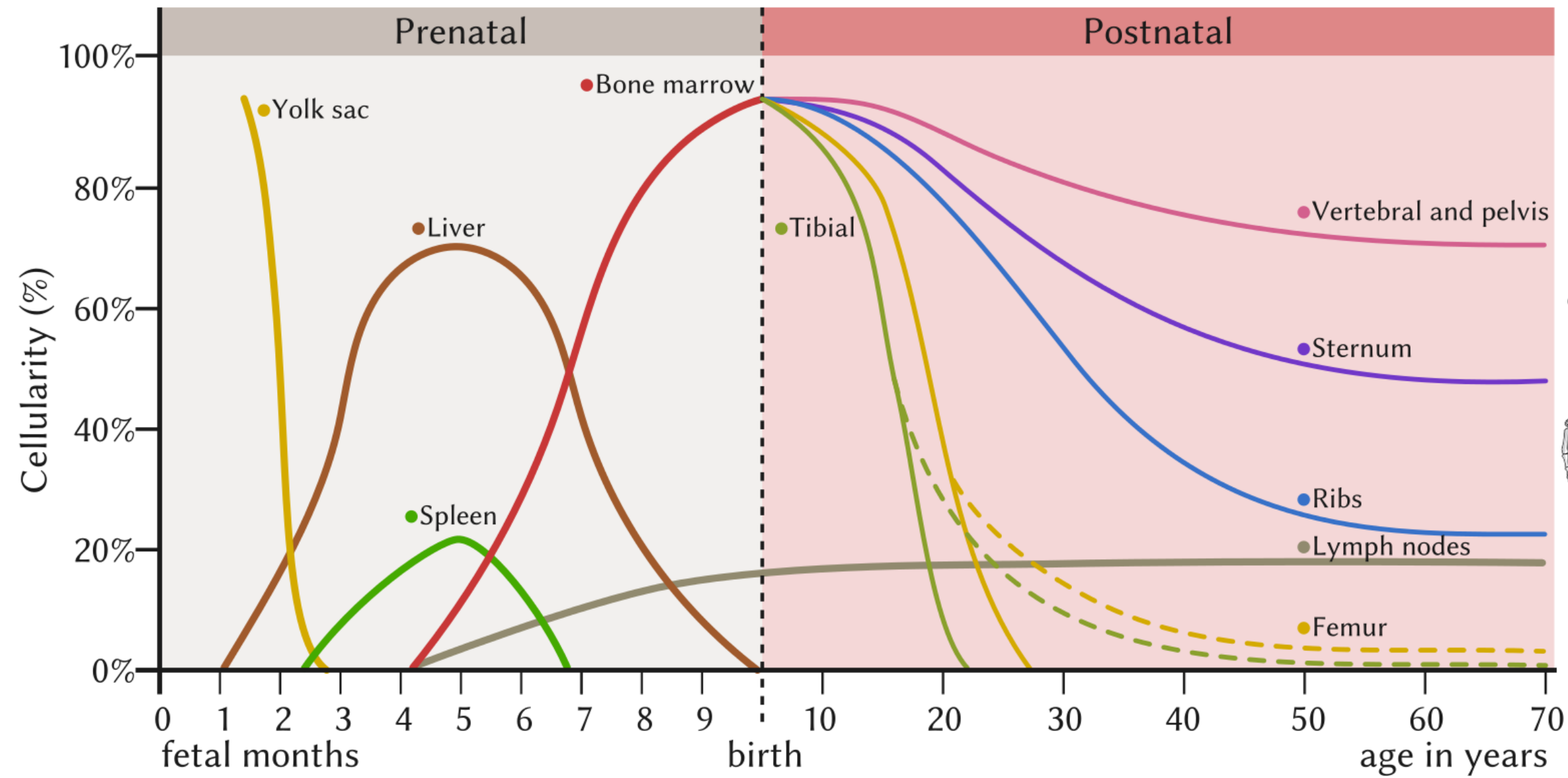


Hematopoiesis Location

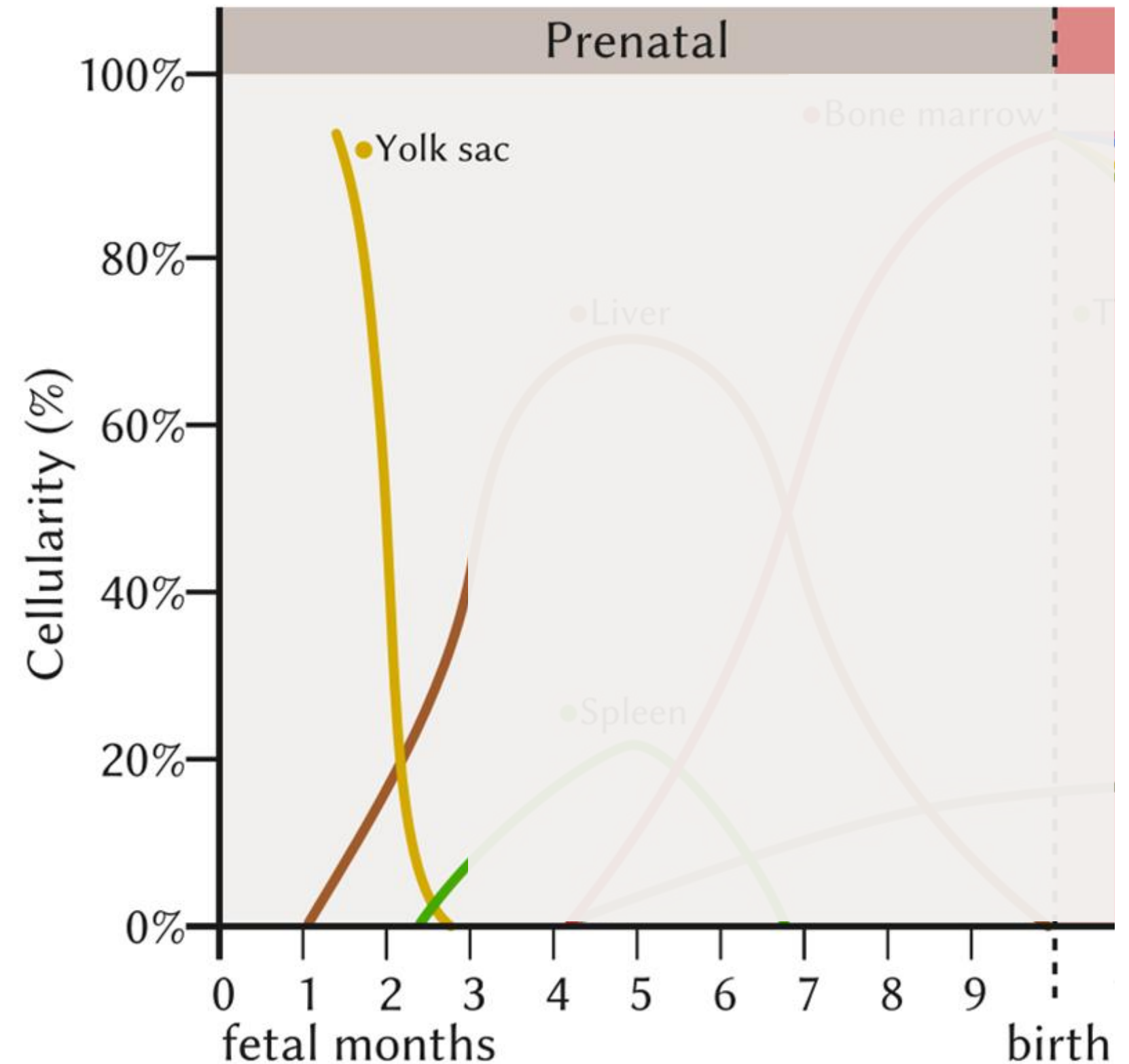
Hematopoiesis



Hematopoiesis

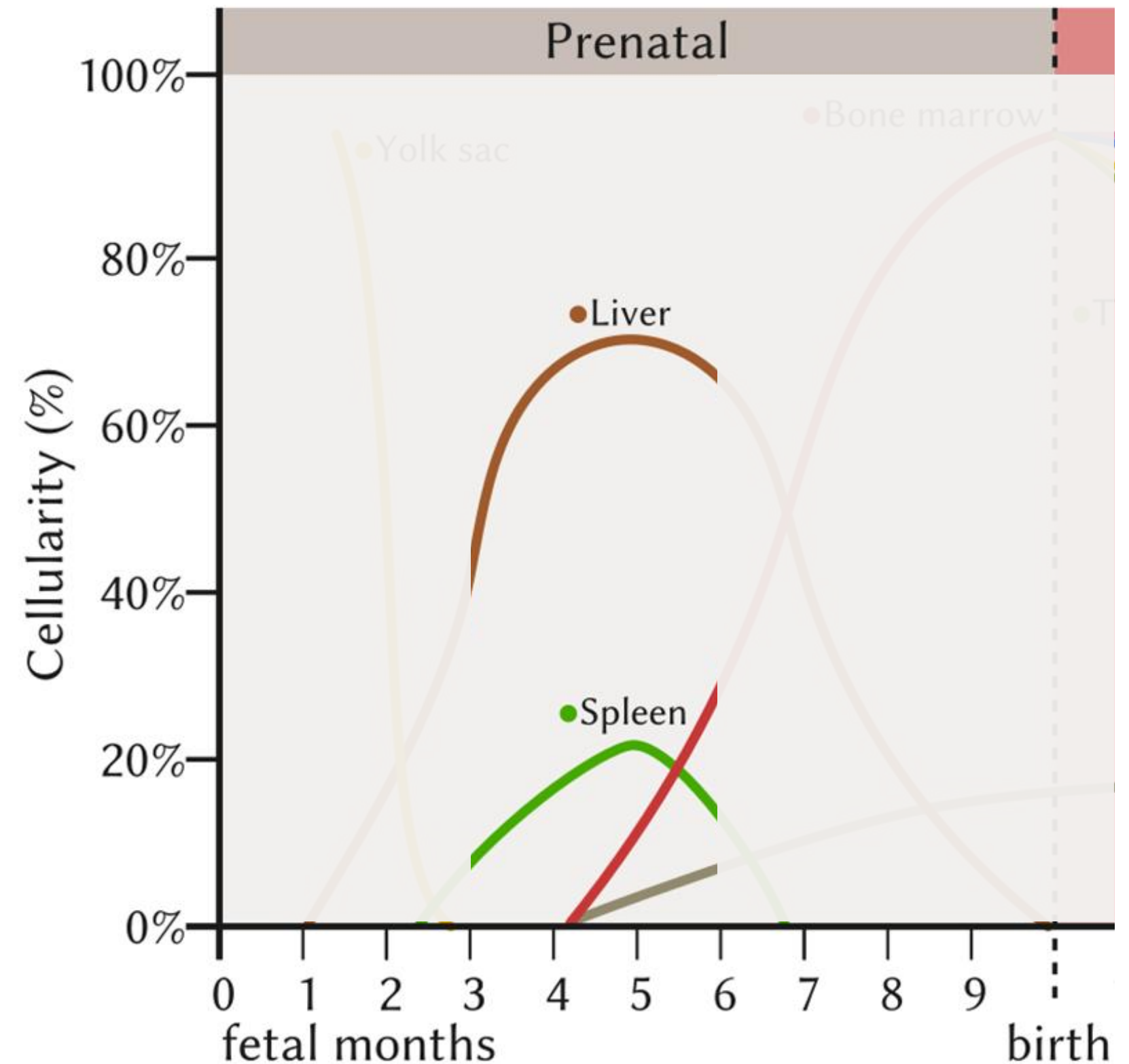
18 days – 3 months

- Yolk sac
 - Erythrocytes
 - Macrophages
 - Primitive erythropoiesis
 - Not from HSCs
- Aorta-gonads-mesonephros
 - Definitive erythropoiesis
 - Begins HSC production



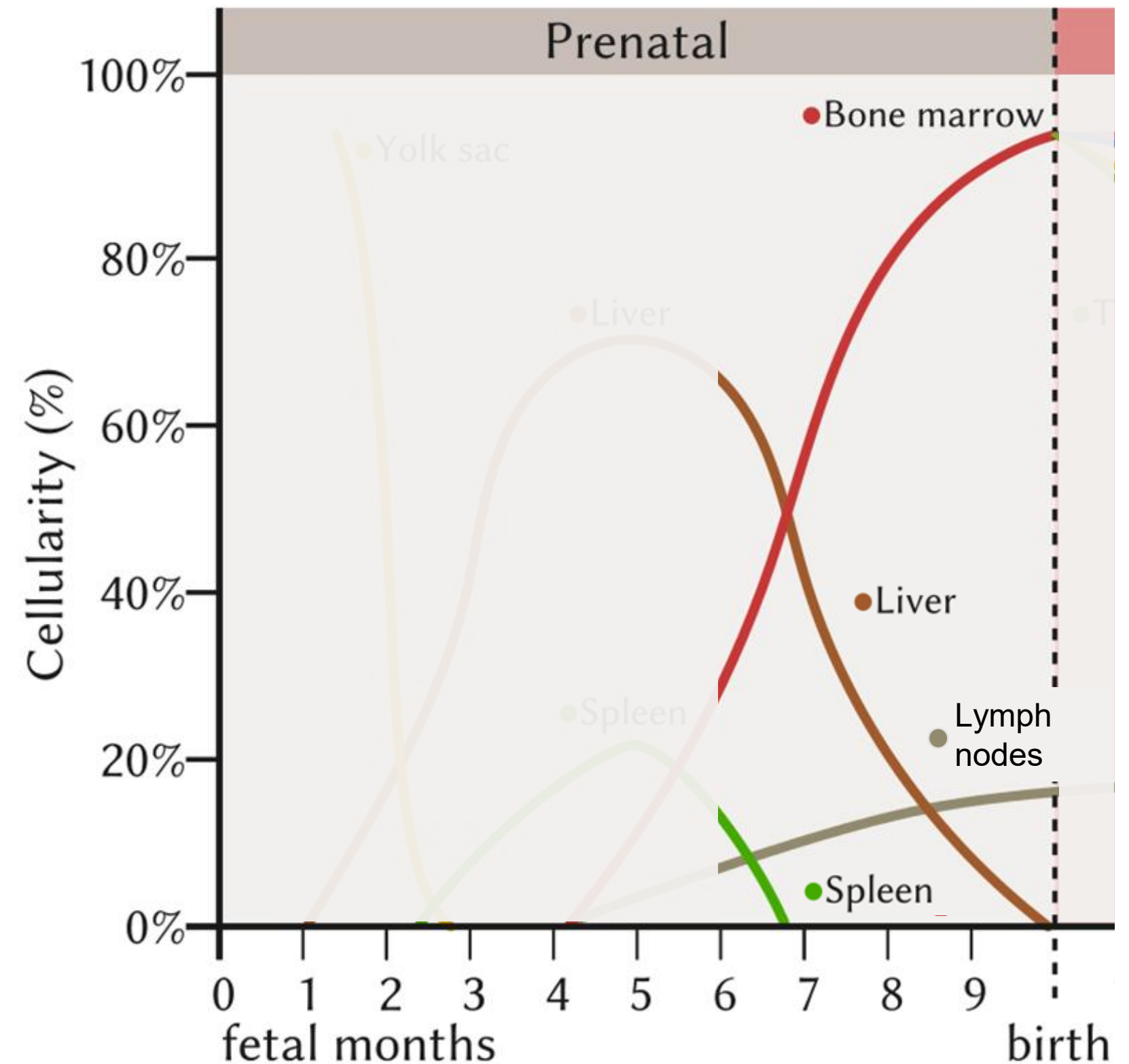
Hematopoiesis

- 3-6 months
- Liver
 - Majority erythrocytes
 - Myeloids
 - Lymphoids
- Other sites
 - Spleen, kidney, thymus, lymph nodes



Hematopoiesis

- 6 months - birth
- Bone Marrow
 - Primary hematopoiesis site
- Thymus
 - Fetal T cells (genetically different from adult T cells)
- Spleen, Lymph nodes
 - B cell differentiation



Hematopoiesis

> birth

1. All cells development

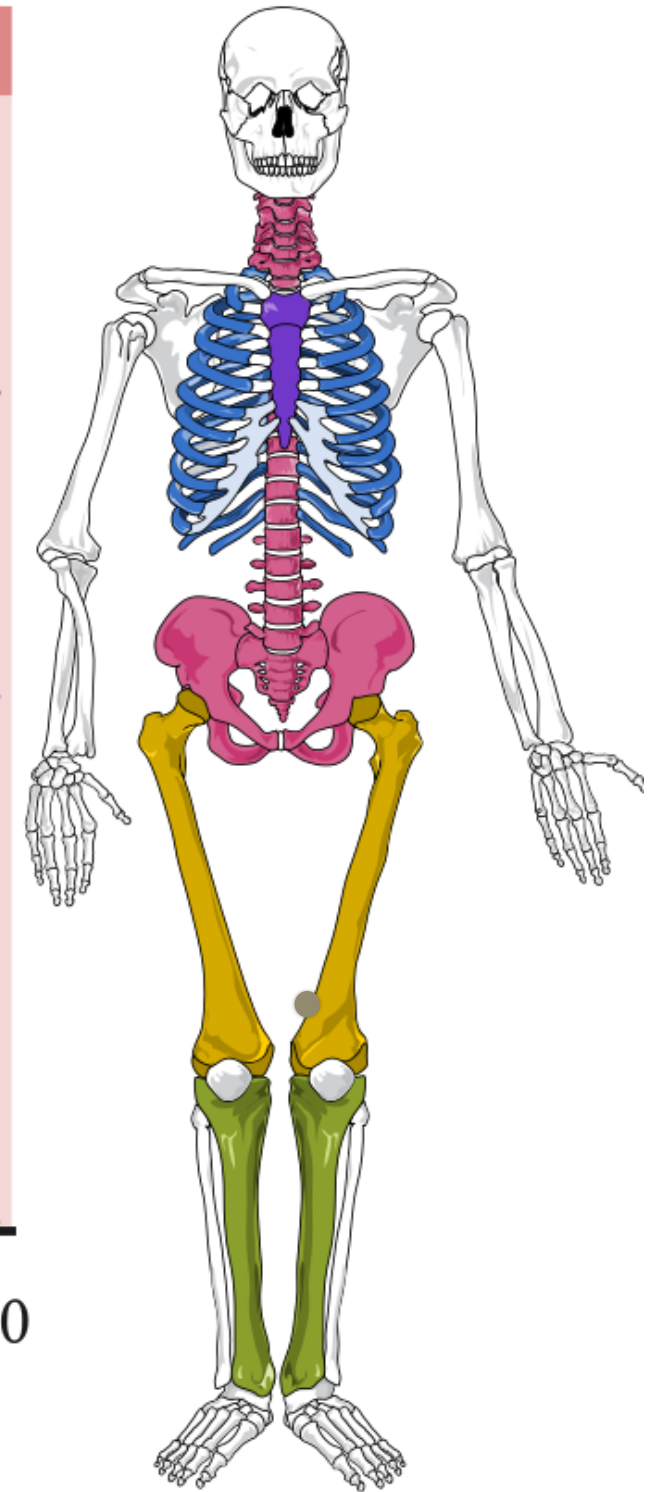
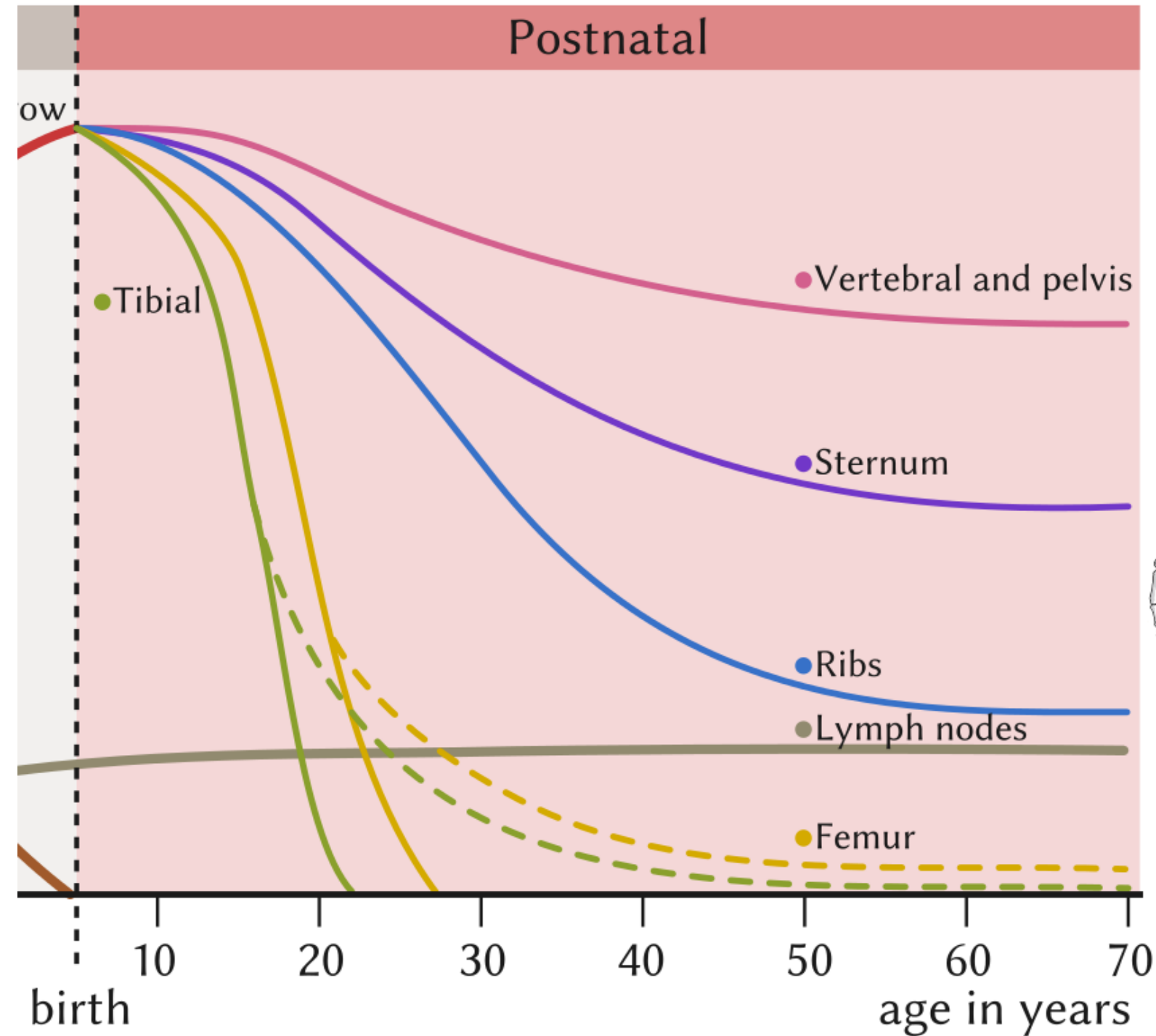
*Bone Marrow

2. Later lymphoid development

**Thymus

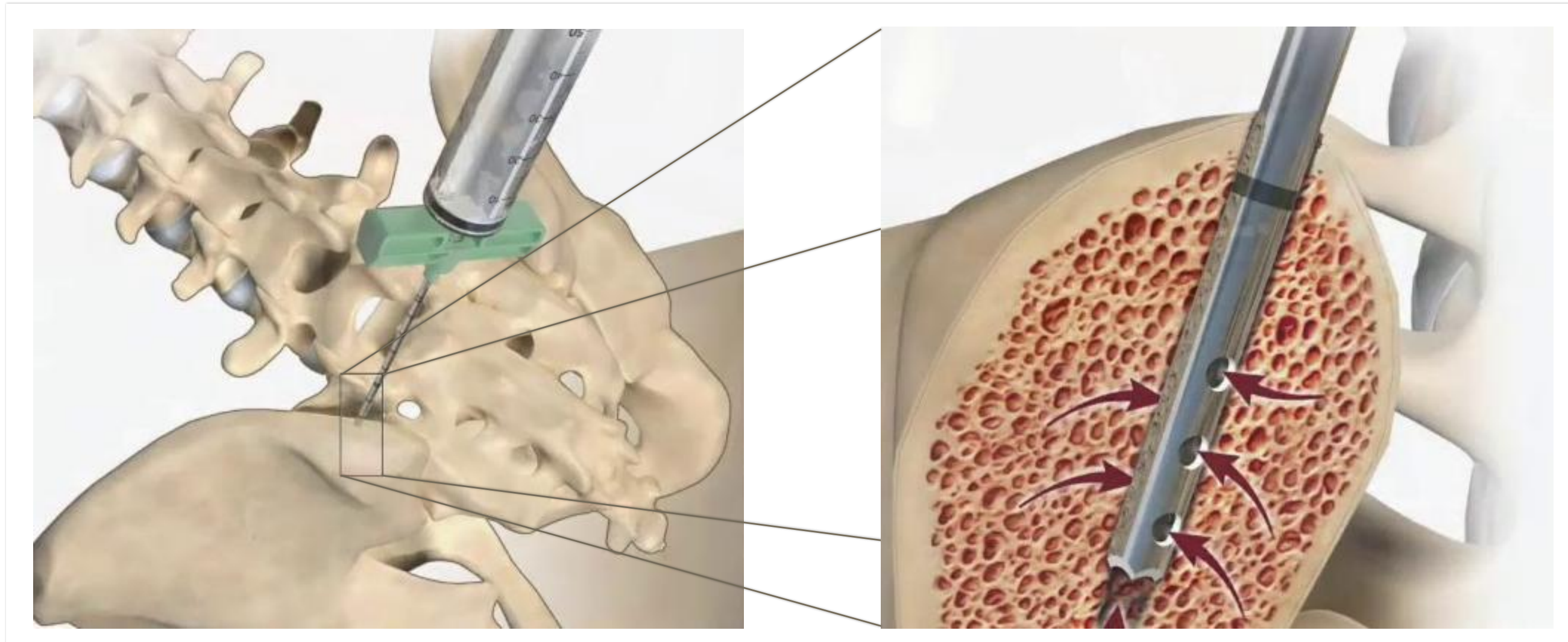
** Spleen

** Lymph nodes



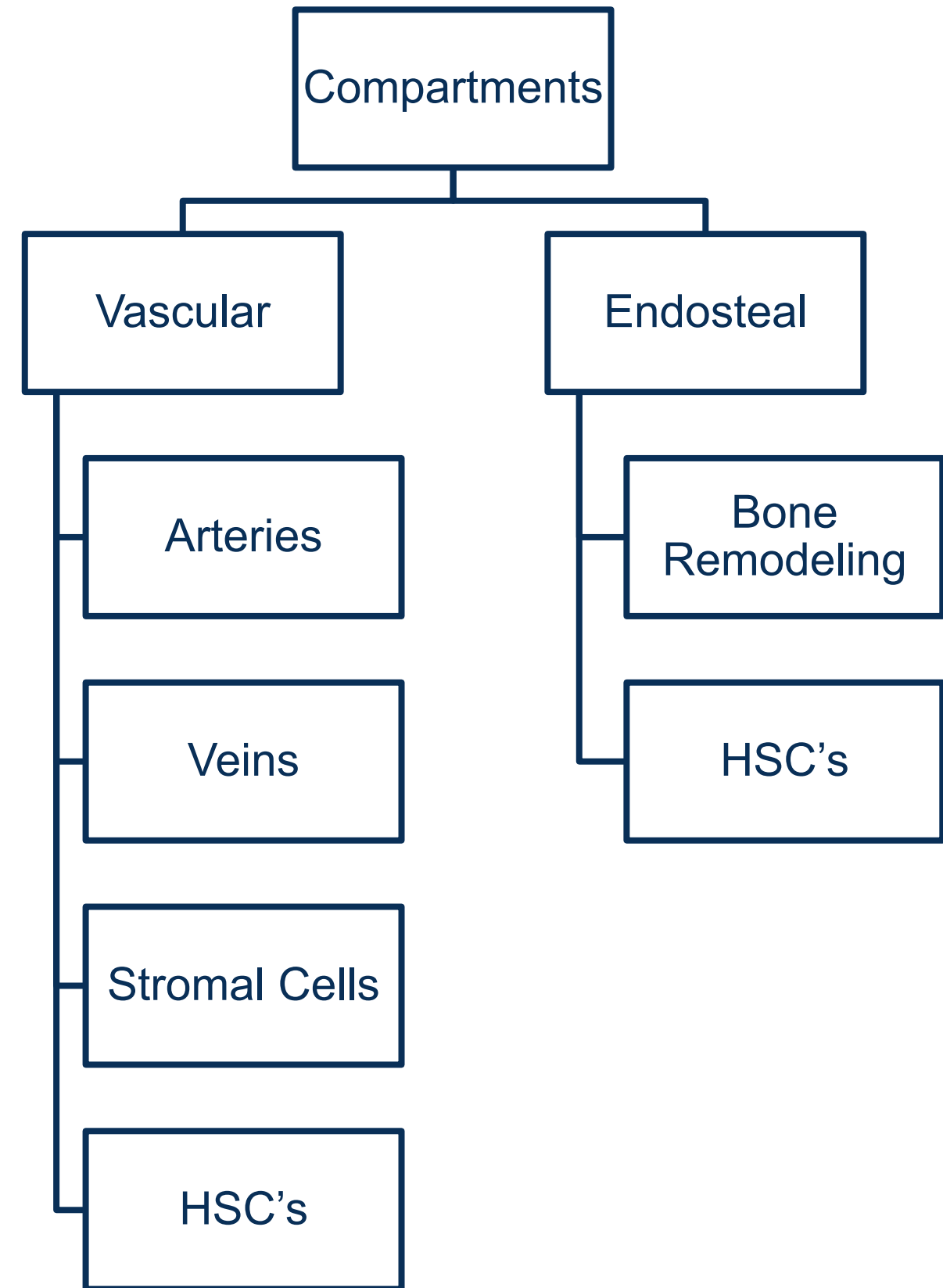
Bone Marrow

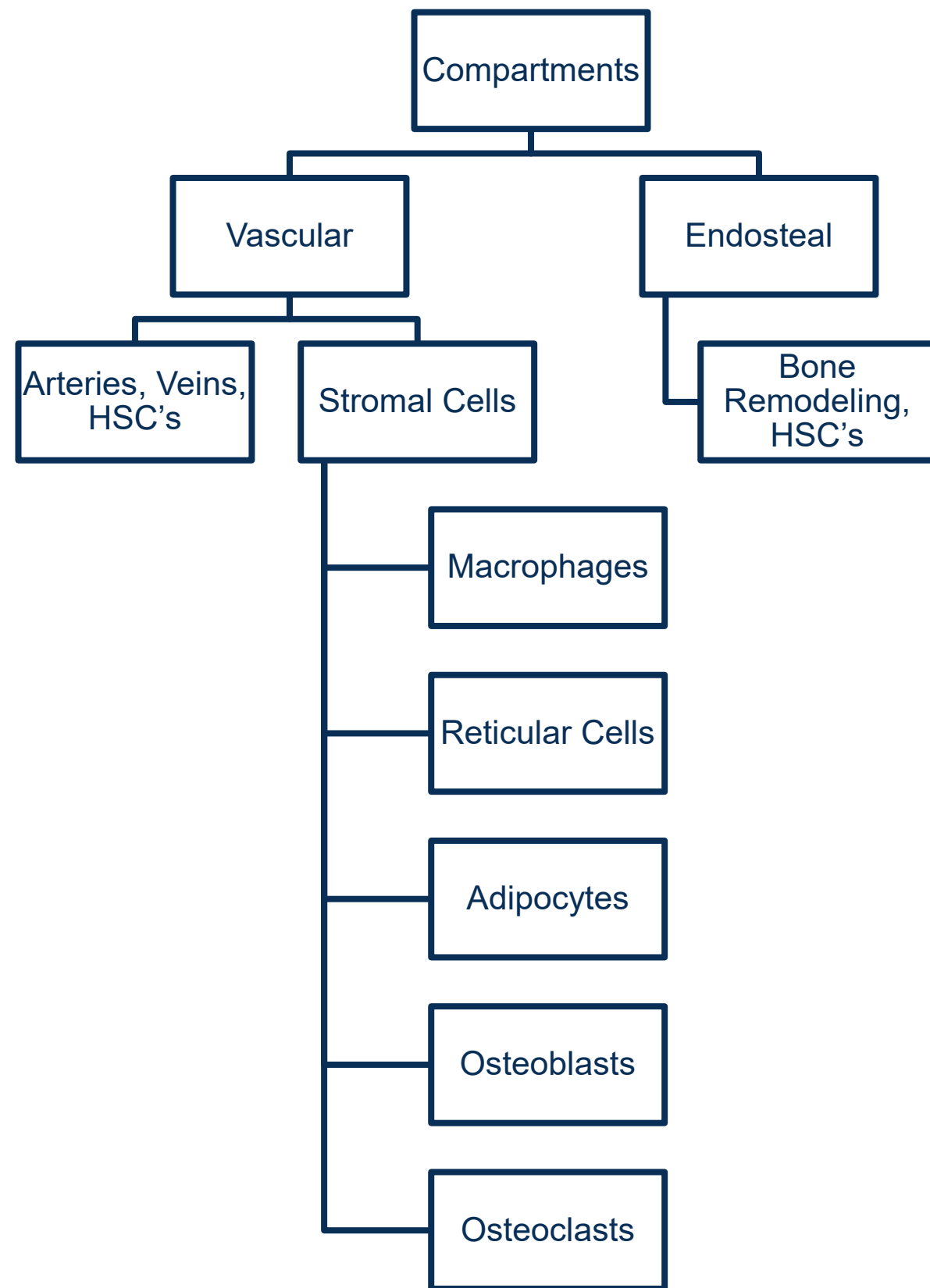
- Blood forming tissue between trabeculae of spongy bone



Bone Marrow

- Blood forming tissue within spongy bone
 - Vascular compartment
 - Arteries and veins
 - Stromal cells
 - Contains hematopoietic stem cells (HSC)
 - Endosteal compartment
 - Contains HSC's

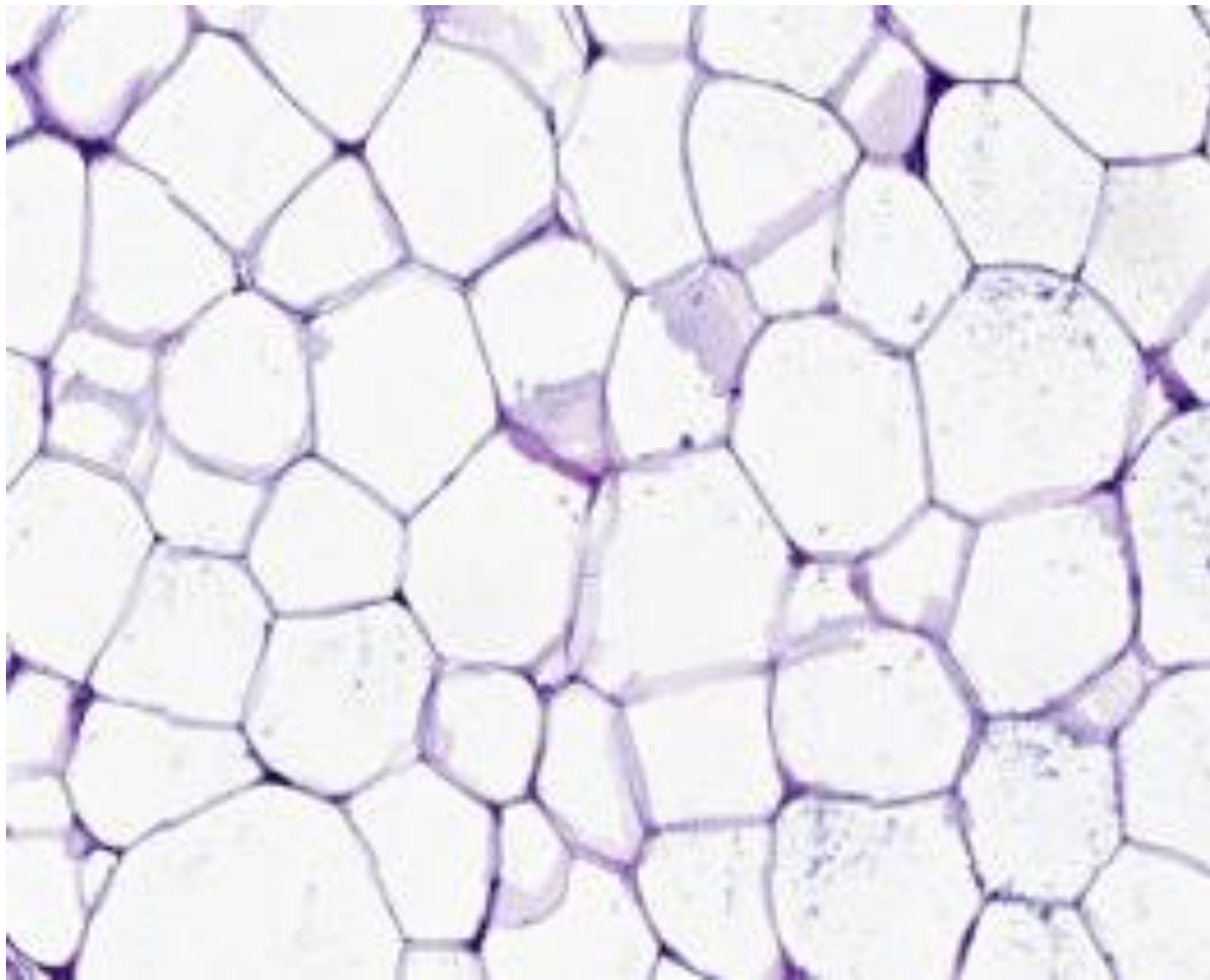




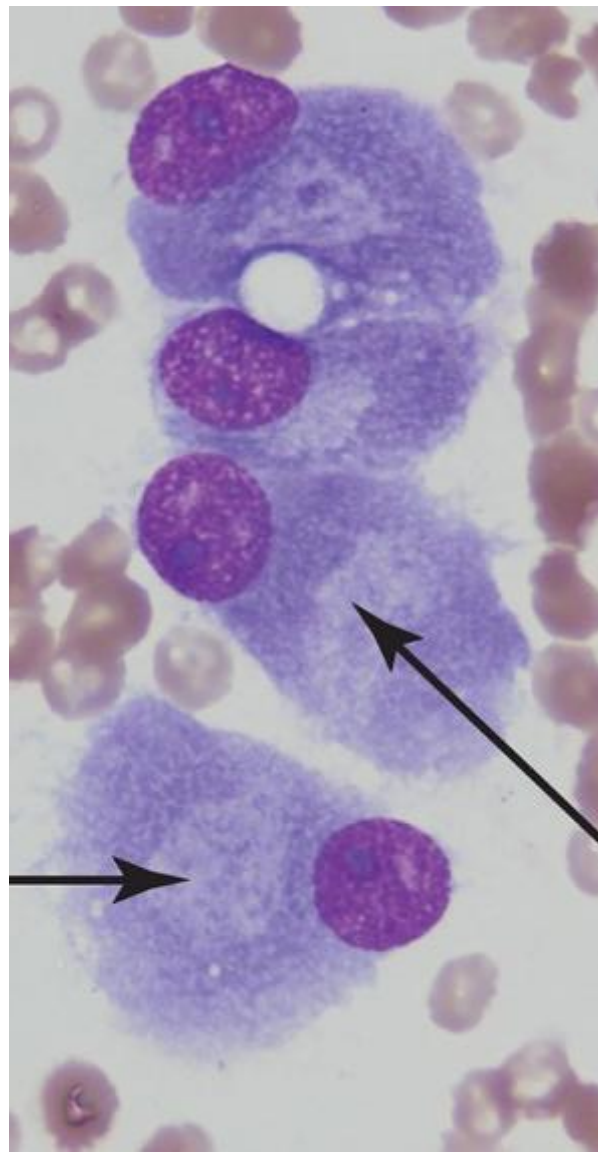
Stromal Cells

- All secrete cytokines to maintain HSC pool
- 3D meshwork provides scaffolding support for bone marrow, composed of:
 - Macrophages – phagocytosis, cytokine secretion
 - Reticular cells – fibroblasts (3D supporting network)
 - Adipocytes – fat storage (volume control)
 - Osteoblast – bone building
 - Osteoclast – bone recycling

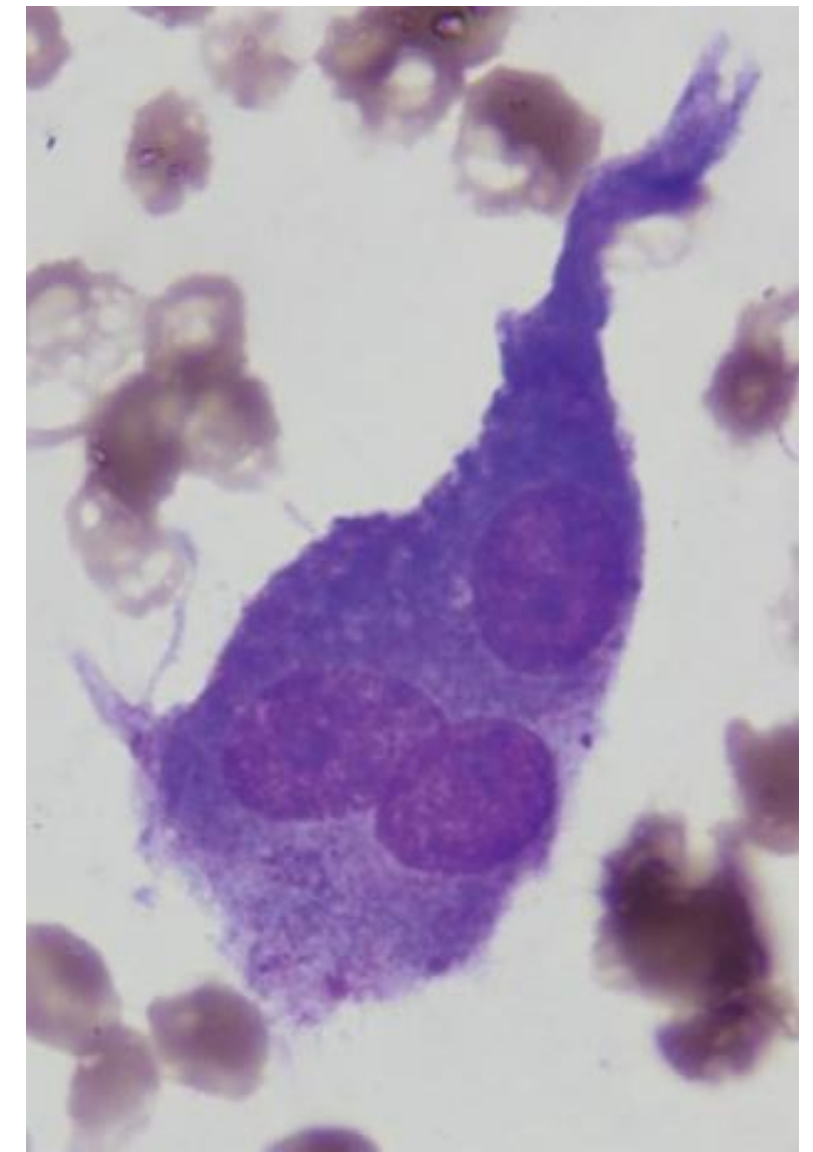
Adipocytes



Osteoblast

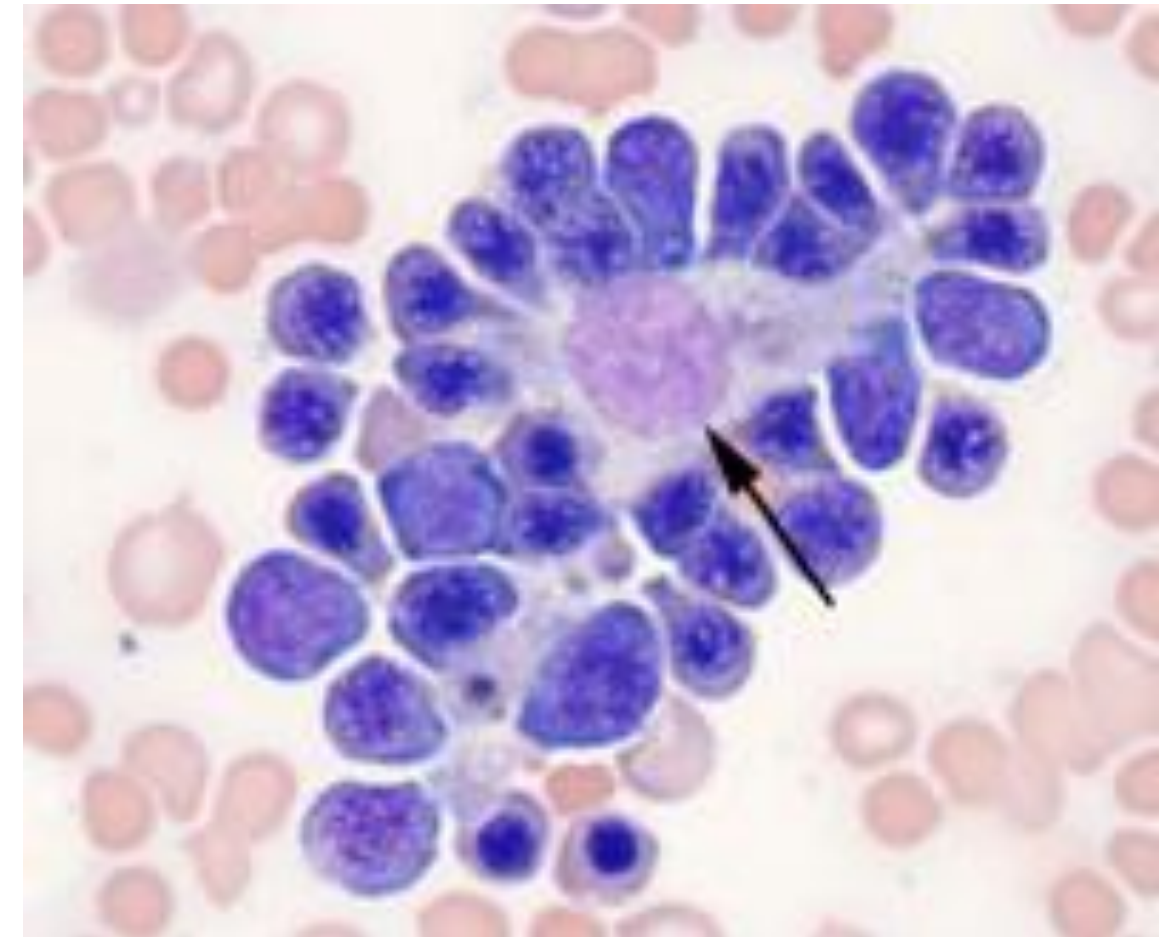
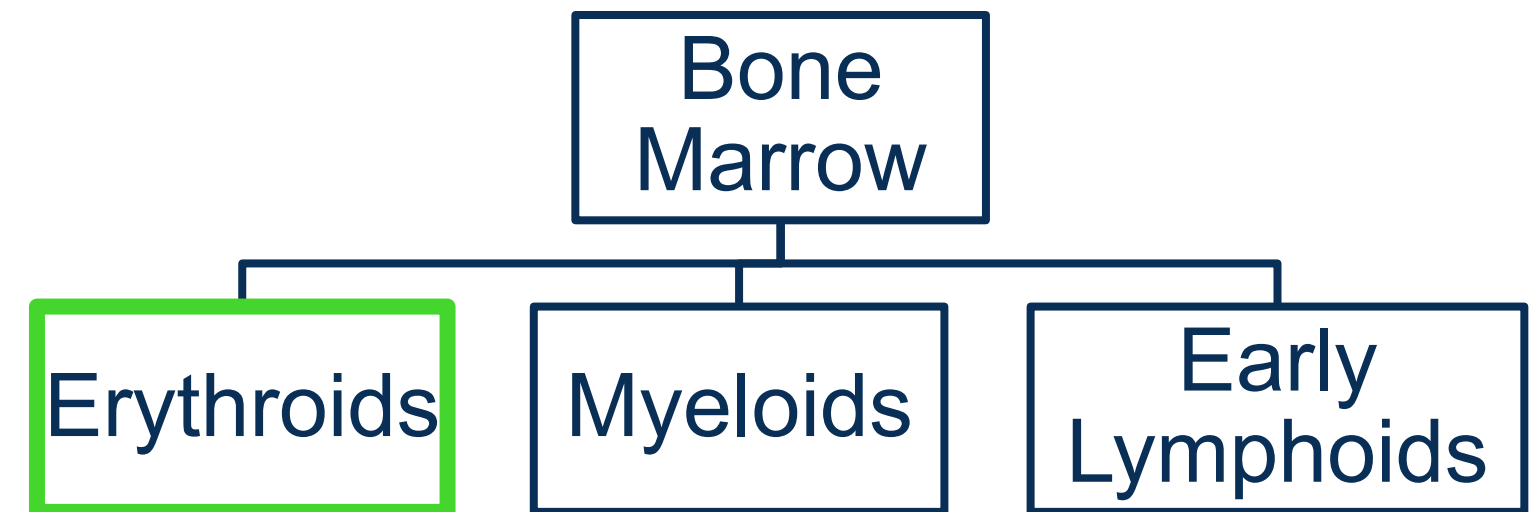


Osteoclast



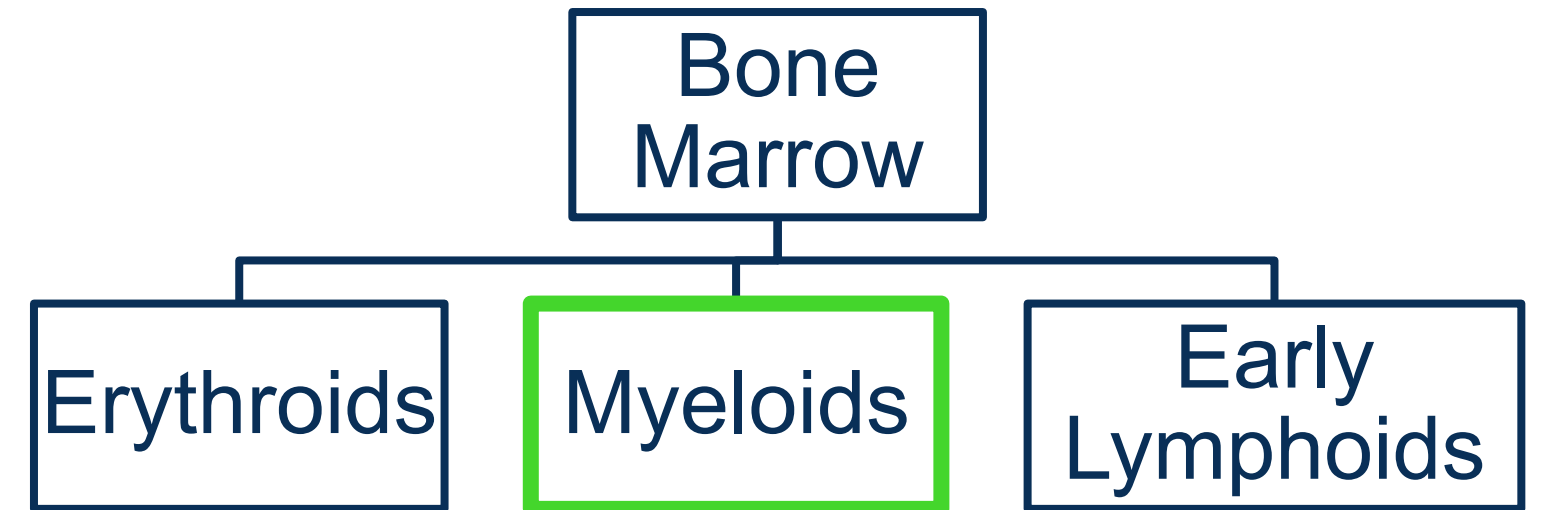
Hematopoietic Cells

- Arranged in niches within marrow cavity
- Erythroids
 - 25-30% of marrow cells
 - Produced near venous sinuses
 - Develop in erythroblastic islands
 - Macrophage regulates via cytokines



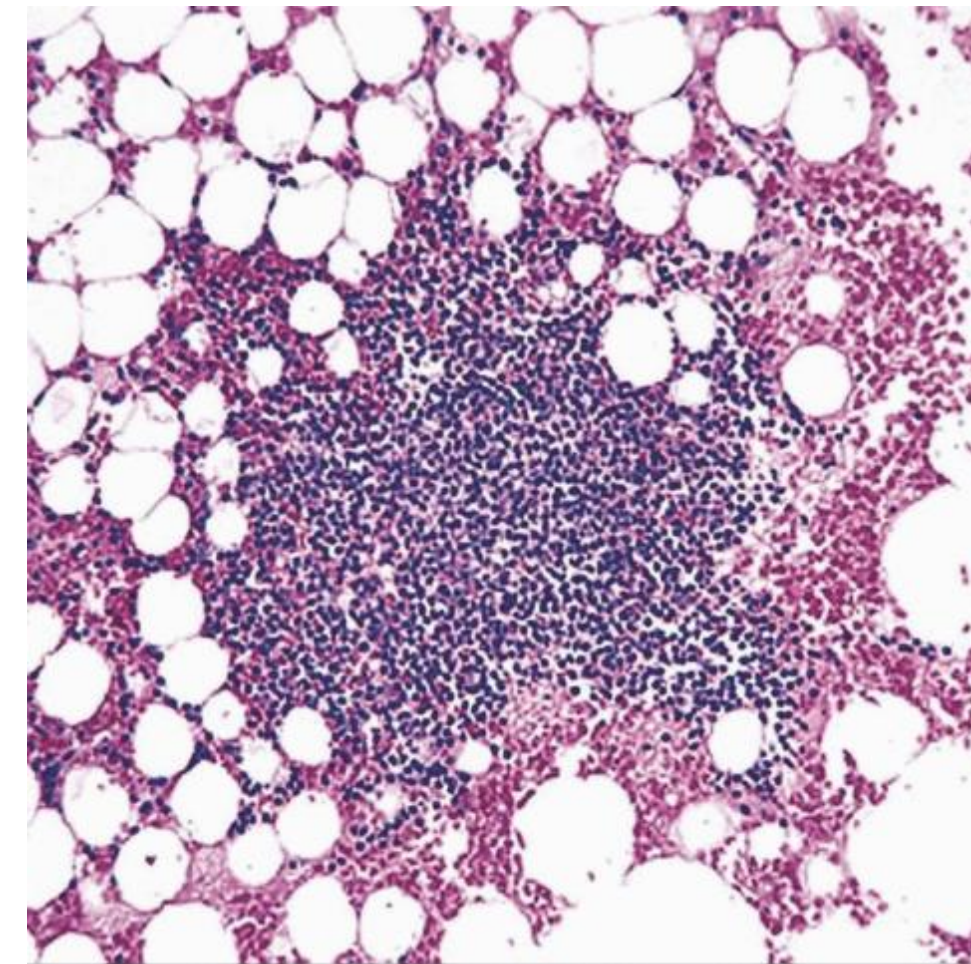
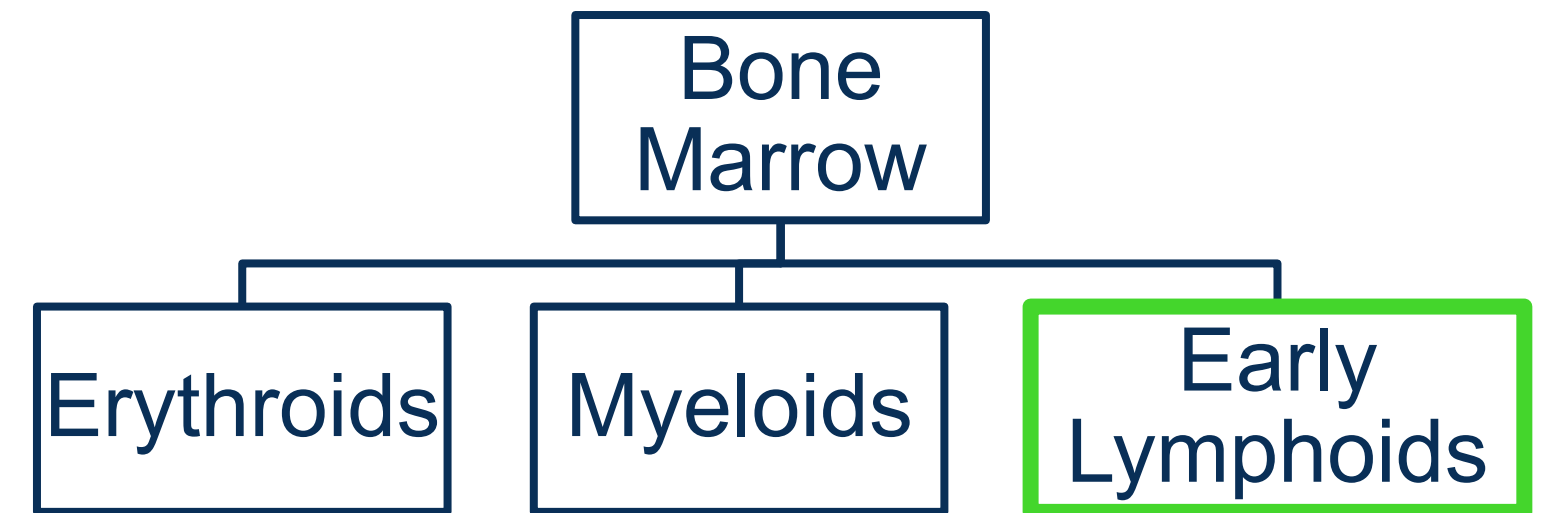
Hematopoietic Cells

- Arranged in niches within marrow cavity
- Myeloids
 - Produced in nests
 - Close to trabeculae and arterioles
 - Distant from venous sinuses



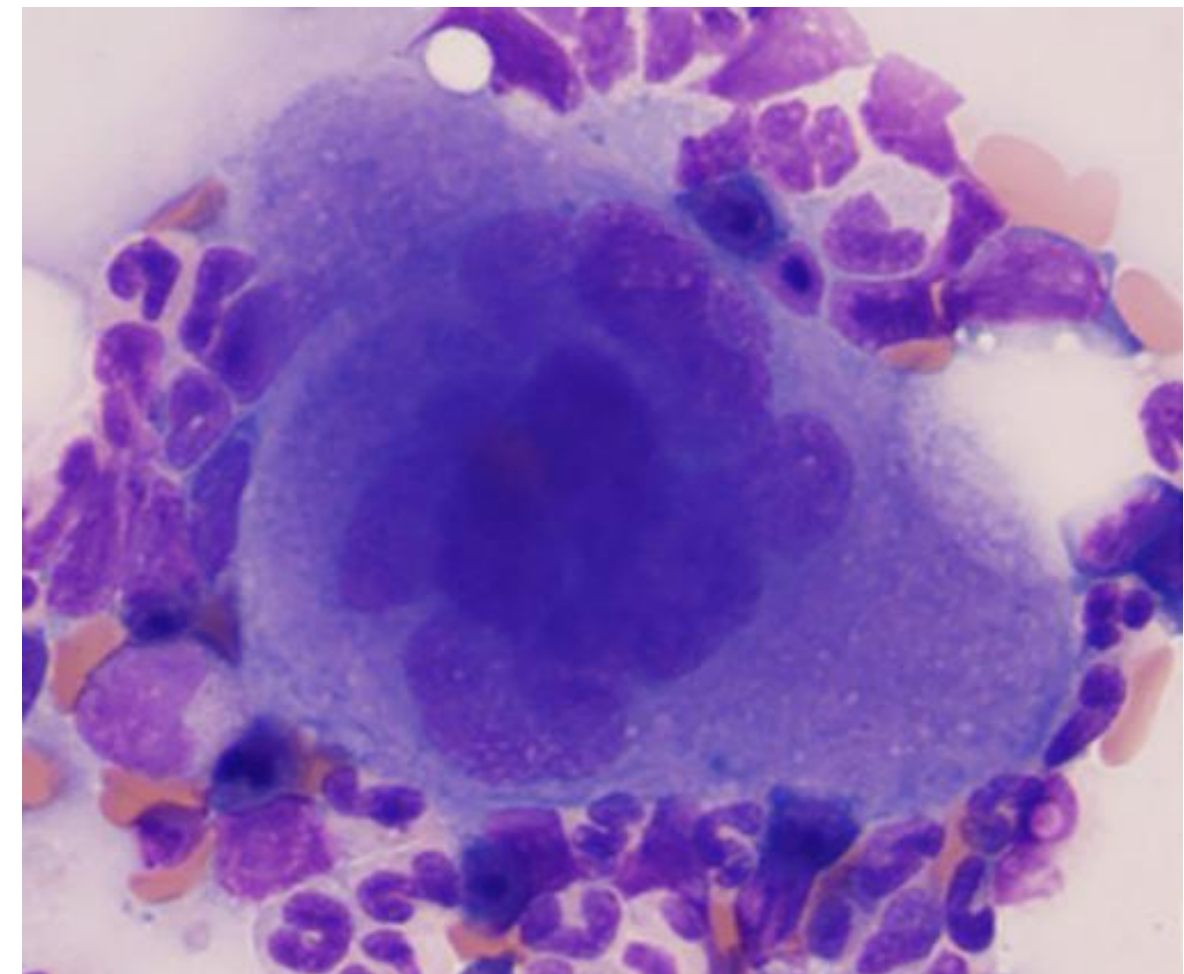
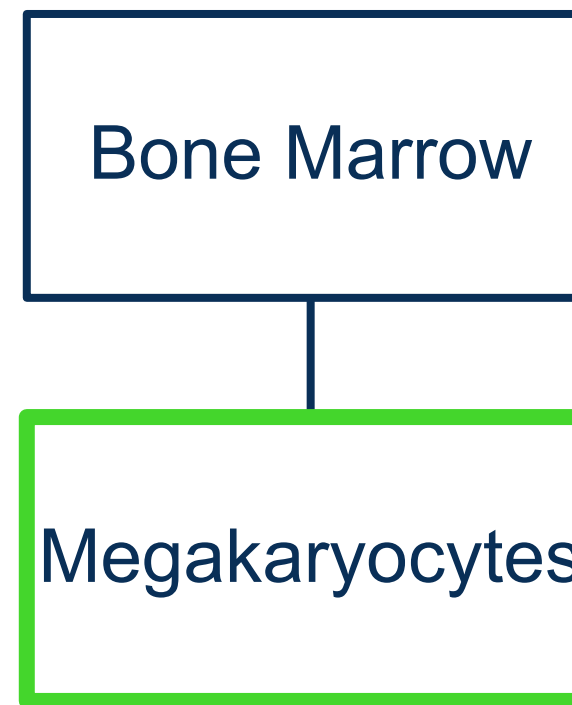
Hematopoietic Cells

- Arranged in niches within marrow cavity
- Lymphoids
- Near arterioles in lymphoid aggregates
- Some progenitors go to
 - Thymus → become T cells
 - Stay in BM → mature into B cells
 - Activated B cells return → transform into plasma cells (antibodies)



Hematopoietic Cells

- Arranged in niches within marrow cavity
- Megakaryocytes
 - Large, polyploidy
 - Cytoplasm fragments to form platelets
 - Near vascular sinus



Cell Egress

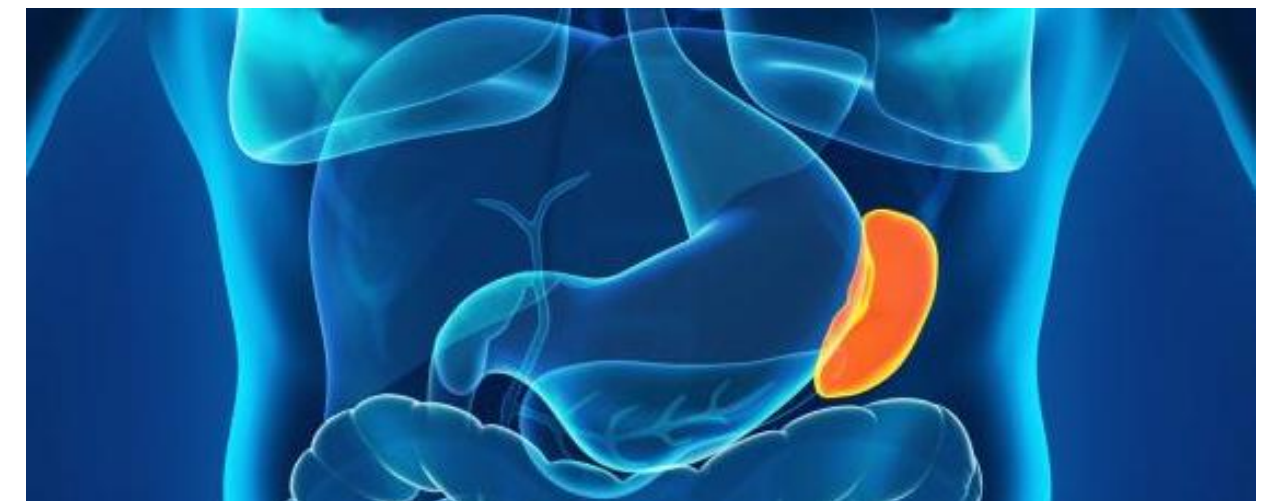
- Soluble factors regulate release of blood cells from bone marrow
 - G-CSF
 - Granulocyte-colony stimulating factor
 - GM-CSF
 - Granulocyte/monocyte-colony stimulating factor
 - Chemokines

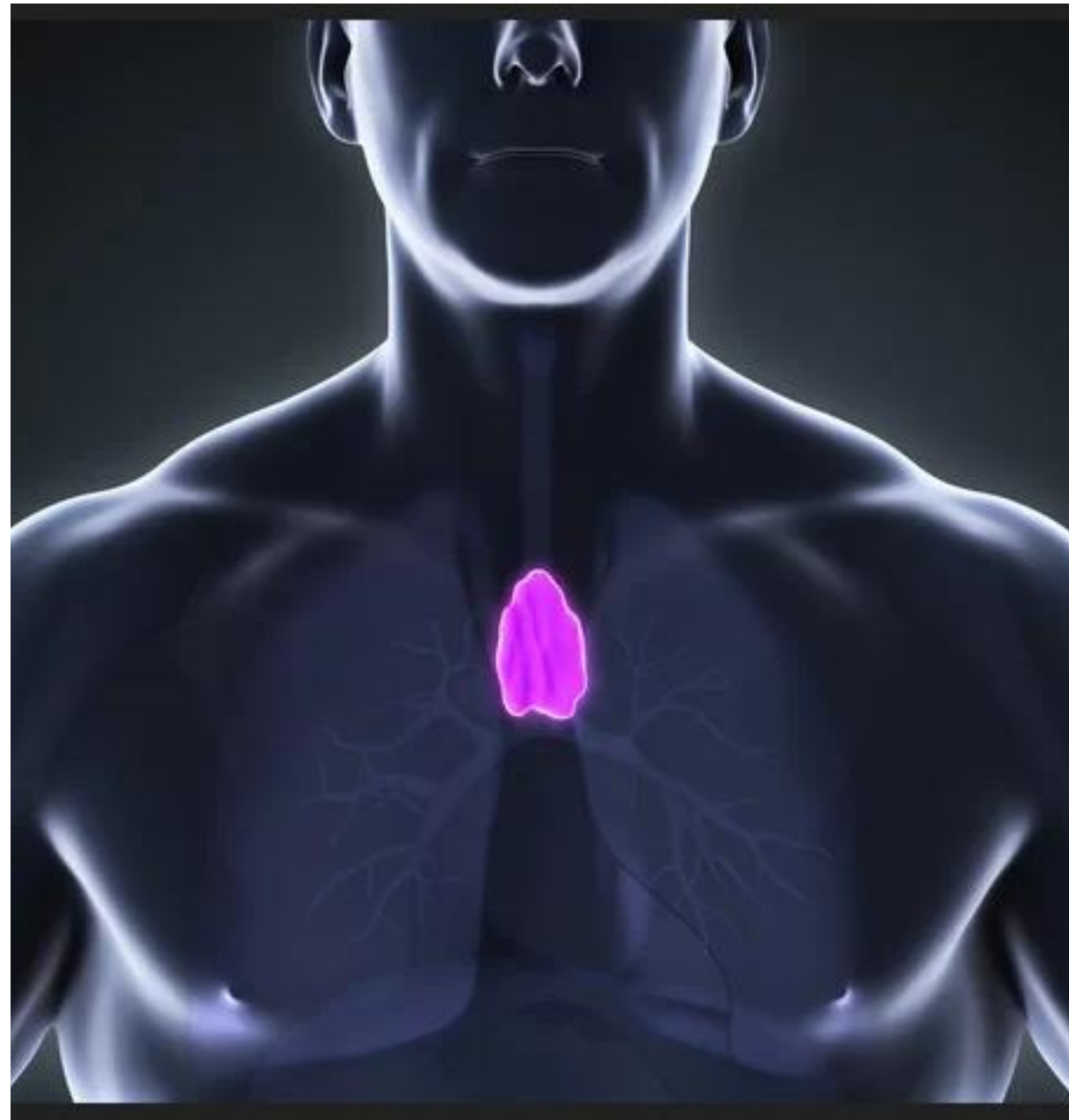
Cell Concentration

- In bone marrow
 - Adipocytes = age $\pm$ 10%
 - Other cells = $100 - (\text{age} \pm 10\%)$
 - Hyperplasia
 - Excess of normal cell production
 - Acute blood loss, chronic anemia, leukemia
 - Hypoplasia
 - Decrease in cell production
 - Chemical exposure, toxins, genetics

Site of Hematopoiesis

- Medullary hematopoiesis
 - Blood cell production in the bone marrow
- Extramedullary hematopoiesis
 - Blood cell production other than bone marrow
 - Liver, spleen
 - Occurs when BM can't meet physiologic need





Thymus

- Lymphopoietic organ
- Well developed at birth
 - Atrophy begins after puberty
- Maturation of T lymphocytes
- Supplies immunocompetent T cells
 - Lymph nodes
 - Spleen
 - Peripheral lymphoid tissue

Spleen

- Fast and slow blood transit pathways
- Slow pathway (reticuloendothelial system, RES)
 - Filters
 - Foreign substances
 - Culling and pitting RBCs
 - Stores 1/3 platelet mass
 - Assists immune defense
 - Initiates response



Spleen

- Splenomegaly
 - Hypersplenism
 - Exaggeration of normal activities
 - Anemia, leukopenia, thrombocytopenia, cytopenias
- Splenectomy
 - Liver assumes function
 - Autosplenectomy – complication of SC
 - Acidic, hypoxic, and hypoglycemic environment
 - => hyposplenism



Lymph Nodes

- Filters foreign particles from lymph [fluid]
- Provide immune defense against pathogens in all tissues
- Lymphadenopathy
 - Enlargement of lymph node(s)
 - Causes
 - Inflammation
 - Prolonged immune response
 - Malignant transformation

